

The Pythagorean Compendium: Universal Edition

Integrative Logic of the Quadrivium and Figurate Number Theory

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Abstract

The Pythagorean system represents the first unified field theory of Western history, asserting that the structure of the cosmos is essentially mathematical. This edition expands the previous specification with the logic of Gnomonic expansion, the formal proof of triangular angularity, and the architectural application of harmonic proportions.

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1 The Quadrivium: The Fourfold Path

The Pythagoreans viewed the universe through four mathematical lenses. These disciplines formed the foundation of the *Artes Liberales*:

1. **Arithmetic:** Discrete quantity at rest (Number).
2. **Music:** Discrete quantity in motion (Ratio and Harmony).
3. **Geometry:** Continuous magnitude at rest (Space).
4. **Astronomy:** Continuous magnitude in motion (Cosmic Order).

2 Geometric and Trigonometric Formalisms

2.1 Triangle Properties and Angles

The Pythagoreans were the first to demonstrate that the internal angles of any triangle are equivalent to two right angles.

$$\angle A + \angle B + \angle C = 180^\circ \quad (\pi \text{ radians}) \quad (1)$$

This was proven via the construction of a line parallel to the base passing through the vertex, utilizing the properties of alternate interior angles.

2.2 Altitude and Reciprocals

In a right triangle with legs a, b , hypotenuse c , and altitude h dropped to the hypotenuse:

$$Area = \frac{1}{2}ab = \frac{1}{2}ch \implies h = \frac{ab}{c} \quad (2)$$

$$\frac{1}{h^2} = \frac{c^2}{a^2b^2} = \frac{a^2 + b^2}{a^2b^2} = \frac{1}{a^2} + \frac{1}{b^2} \quad (3)$$

3 Figurate Number Theory and the Gnomon

3.1 The Gnomon Method

A "Gnomon" (meaning "that which knows" or "indicator") is the shape added to a figure to produce a new figure of the same type.

- **Square Numbers:** The gnomon of a square number n^2 is the n -th odd number $(2n + 1)$.

$$1 \xrightarrow{+3} 4 \xrightarrow{+5} 9 \xrightarrow{+7} 16 \dots$$

- **Oblong Numbers:** Numbers of the form $n(n + 1)$. The sum of the first n even numbers.

3.2 Polygonal Summation

The n -th k -gonal number $P(k, n)$ can be calculated using the formula:

$$P(k, n) = \frac{(k - 2)n^2 - (k - 4)n}{2} \quad (4)$$

For the Tetractys (Triangular, $k = 3$): $P(3, n) = \frac{n^2 + n}{2}$.

4 Harmonic Theory of the Monochord

4.1 Ratios of String Length

By dividing a single string (the monochord) at specific integer intervals, the school defined the harmonic series:

Length Ratio	Frequency Ratio	Musical Interval
1:1	1:1	Unison
1:2	2:1	Octave
2:3	3:2	Perfect Fifth
3:4	4:3	Perfect Fourth
8:9	9:8	Whole Tone

4.2 The Spiral of Fifths

The "Pythagorean Tuning" system builds an entire scale by stacking Perfect Fifths (3:2). However, this creates the **Pythagorean Comma**:

$$\text{Comma} = \frac{(3/2)^{12}}{2^7} = \frac{531441}{524288} \approx 23.46 \text{ cents} \quad (5)$$

5 Architectural and Physical Applications

5.1 The 3-4-5 Triangle in Practice

Used by the "Harpenodaptai" (rope-stretchers) of Egypt and formalized by Pythagoreans, this triangle ($3^2 + 4^2 = 5^2$) allowed for the perfect squaring of temple foundations and grid-based city planning in Miletus and Thurii.

5.2 Therapeutic Music

The Pythagoreans utilized specific *Modes* to influence the "tuning" of the human soul.

- **Dorian:** Stability and courage.
- **Phrygian:** Excitement and passion.

6 Symbolic Metadata and Constants

- **Pentagram (ϕ):** The ratio of a diagonal to a side in a regular pentagon is exactly $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$.
- **The Tetrad (4):** Represents the four elements (Earth, Air, Fire, Water) and the four directions.